

Classification and Predictive Performance

Alexander McLain

June 5, 2026

Outline

Quick Recap: From Logistic Regression to Probabilities

From Probabilities to Classifications

Sensitivity, Specificity, and the Confusion Matrix

ROC Curves and AUC

Choosing a Threshold

Caveats

Review

Where We Left Off

- ▶ Yesterday we fit logistic regression models to predict the probability of CHD in the Framingham data.
- ▶ Each participant now has a predicted probability $\hat{\pi}_i$ from a fitted model.
- ▶ Today's question: once we have those predicted probabilities, **what do we do with them?**
 - ▶ How do we turn them into yes/no decisions?
 - ▶ How do we evaluate whether those decisions are any good?
 - ▶ How do we compare two competing models?
- ▶ These questions apply to *any* model that produces predicted probabilities, not just logistic regression: random forests, neural networks, gradient boosting, etc.

Choosing a Threshold

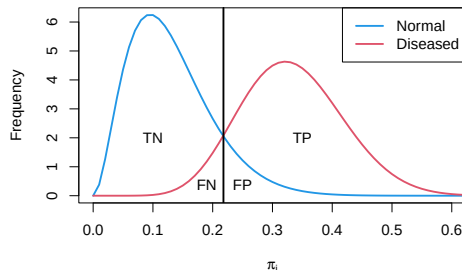
- ▶ Logistic regression returns $\hat{\pi}_i$, an estimated probability for each person.
- ▶ Often we want a hard decision: classify person i as positive ($\hat{Y}_i = 1$) or negative ($\hat{Y}_i = 0$).
- ▶ Pick a threshold π_0 and classify:

$$\hat{Y}_i = \begin{cases} 1 & \text{if } \hat{\pi}_i > \pi_0 \\ 0 & \text{if } \hat{\pi}_i \leq \pi_0 \end{cases}$$

- ▶ $\pi_0 = 0.5$ is the conventional default but rarely the best choice. In Framingham only 15% develop CHD, so few people will have $\hat{\pi}_i$ above 0.5. A lower threshold may be more useful.
- ▶ The right threshold depends on the relative cost of false positives versus false negatives in your application.

Four Things That Can Happen

- ▶ Suppose we use a logistic regression score to screen for a disease. Four things can happen for each person:
 - ▶ **True positive (TP)**: sick person, correctly flagged.
 - ▶ **False positive (FP)**: healthy person, incorrectly flagged.
 - ▶ **True negative (TN)**: healthy person, correctly cleared.
 - ▶ **False negative (FN)**: sick person, missed.
- ▶ A useful classifier produces many TPs and TNs, and few FPs and FNs. But there is always a tradeoff.



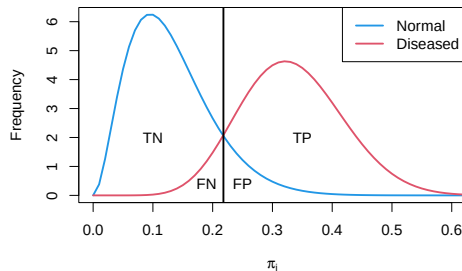
The Confusion Matrix

	Test Pos. (T+)	Test Neg. (T-)	Totals
Diseased (D+)	TP	FN	TP+FN
Not diseased (D-)	FP	TN	FP+TN
Totals	TP+FP	FN+TN	n

Every threshold produces a different confusion matrix. Changing the threshold shifts entries between the columns.

Four Key Summary Measures

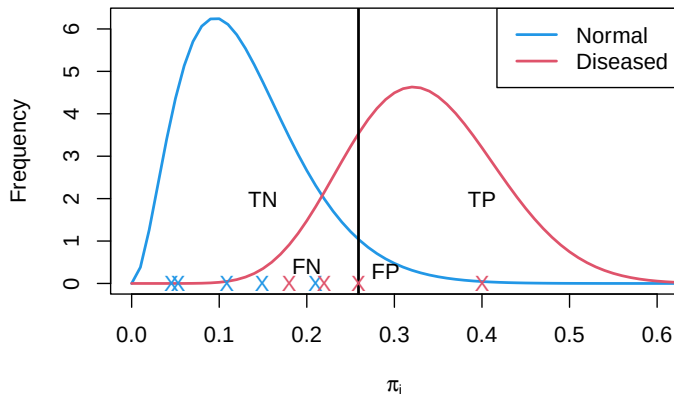
- ▶ **Sensitivity** (true positive rate) = $\frac{TP}{TP + FN}$: of all the truly sick, what fraction did we catch?
- ▶ **Specificity** (true negative rate) = $\frac{TN}{TN + FP}$: of all the truly healthy, what fraction did we correctly clear?
- ▶ **PPV** (positive predictive value) = $\frac{TP}{TP + FP}$: of all the people we flagged, what fraction were really sick?
- ▶ **NPV** (negative predictive value) = $\frac{TN}{TN + FN}$: of all the people we cleared, what fraction were really healthy?



Sensitivity vs. Specificity: The Tradeoff

- ▶ As we raise the classification threshold π_0 :
 - ▶ Fewer people get flagged as positive.
 - ▶ Specificity goes up (we make fewer false alarms).
 - ▶ Sensitivity goes down (we miss more true cases).
- ▶ As we lower π_0 , the tradeoff runs in reverse.
- ▶ Two extreme thresholds:
 - ▶ $\pi_0 = 0$: flag everyone. Sensitivity = 1, specificity = 0.
 - ▶ $\pi_0 = 1$: flag no one. Sensitivity = 0, specificity = 1.
- ▶ Both extremes are useless. A good model makes the middle ground favorable: high sensitivity at almost every level of specificity.

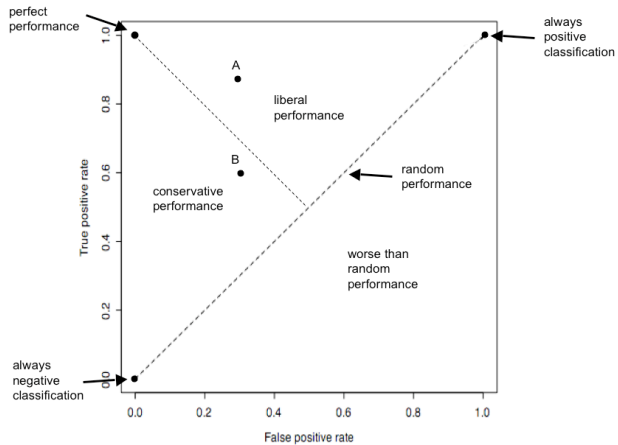
Choosing the Threshold: Visual Intuition



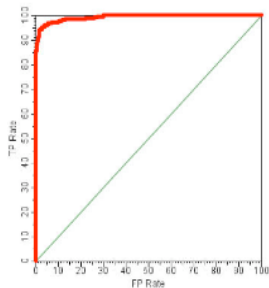
As the threshold slides from left to right, every observation eventually crosses from “positive” to “negative.” The confusion matrix is a snapshot at one threshold; the ROC curve summarizes *every* threshold at once.

Receiver Operating Characteristic (ROC) Curves

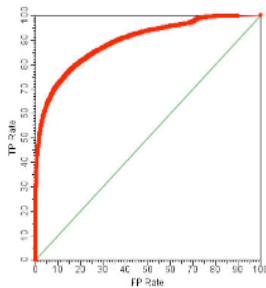
- ▶ An **ROC curve** plots sensitivity against $(1 - \text{specificity})$ as the threshold π_0 varies from 0 to 1.
- ▶ Endpoints of the curve:
 - ▶ $\pi_0 = 0$: top-right corner (everyone flagged: sensitivity 1, $1 - \text{specificity}$ 1).
 - ▶ $\pi_0 = 1$: bottom-left corner (no one flagged: sensitivity 0, $1 - \text{specificity}$ 0).
- ▶ **Diagonal line**: the curve you would get from a coin flip. At every threshold, the test catches sick and healthy people in equal proportions.
- ▶ **Curves that bow up and to the left** represent better classifiers: high sensitivity at low false-positive rates.
- ▶ In R : `pROC::roc(outcome, predicted_probs)` produces the curve and computes the area under it.



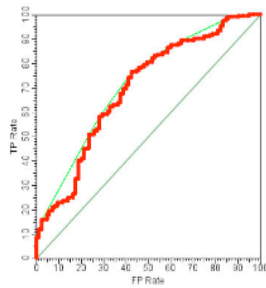
Three ROC Curves Compared



Almost perfect
prediction



Good prediction



Poor prediction

The leftmost curve hugs the top-left corner: high sensitivity even at very high specificity. The rightmost curve barely separates from the diagonal.

Area Under the ROC Curve (AUC)

- ▶ **AUC** is the single most popular summary of an ROC curve. A larger AUC means a better classifier.
- ▶ Concrete interpretation: AUC is the probability that the model assigns a higher predicted probability to a randomly chosen *sick* person than to a randomly chosen *healthy* person.

$$\text{AUC} = P(\hat{\pi}_i > \hat{\pi}_j \mid Y_i = 1, Y_j = 0)$$

- ▶ AUC is threshold-free: it summarizes performance across all possible π_0 .
- ▶ It does not, on its own, tell you which threshold to use. That requires bringing in the cost of false positives vs. false negatives.

Interpreting AUC Values

$\widehat{\text{AUC}}$	Discriminative ability
< 0.5	worse than chance (likely a bug)
$= 0.5$	no discrimination (coin flip)
$0.7 - 0.8$	acceptable discrimination
$0.8 - 0.9$	excellent discrimination
> 0.9	outstanding discrimination

- ▶ These cutoffs are rules of thumb.
- ▶ AUC below 0.6 is rarely useful in clinical settings.
- ▶ The most meaningful comparison is usually the AUC for your model vs. the AUC for a simpler baseline.

The Threshold Is a Clinical Decision

- ▶ The threshold π_0 is a decision about the relative cost of two types of mistakes.
- ▶ **CHD screening:** missing a true case (false negative) means a patient does not get preventive treatment. False positives mean unnecessary follow-up tests. Most screening programs set the threshold low (high sensitivity), accepting many false positives to catch as many true cases as possible.
- ▶ **Costly one-time intervention:** the threshold is set higher, so that only people very likely to benefit are treated.
- ▶ **No threshold is “correct” in the abstract.** The right threshold depends on the clinical context, the cost of each type of error, and the preferences of the patient and clinician.
- ▶ In R , Youden's J statistic (`coords(roc, "best", best.method = "youden")`) gives the threshold maximizing sensitivity + specificity - 1, a reasonable default when costs are unclear.

Training AUC vs. Honest AUC

- ▶ The AUC computed on the same data used to fit the model is **optimistic**. The model has already seen these data, so it looks better than it really is.
- ▶ For an honest estimate:
 - ▶ Split the data into training and test sets. Fit on the training set, evaluate AUC on the test set.
 - ▶ Or use cross-validation (Day 7): rotate the test set across folds and average the AUCs.
- ▶ The gap between training AUC and honest AUC is larger when:
 - ▶ The model has many predictors relative to the sample size.
 - ▶ Variable selection (stepwise, LASSO) was performed on the training data.
- ▶ Always report which evaluation scheme was used. A training AUC of 0.85 and a cross-validated AUC of 0.72 tell very different stories.

Review: Classification and Predictive Performance

- ▶ Predicted probabilities from logistic regression become yes/no classifications once you choose a threshold.
- ▶ The confusion matrix, sensitivity, specificity, PPV, and NPV summarize performance at one threshold.
- ▶ The ROC curve summarizes performance across all thresholds.
- ▶ AUC is a single-number summary of discriminative ability.
- ▶ The threshold is a clinical decision, not a statistical one.
- ▶ Always evaluate performance on held-out data or via cross-validation to avoid optimistic estimates.
- ▶ **Coming up:** survival analysis, where we model not just *whether* an event happened but *when*.